

Accelerated Linear Growth during Erdafitinib Treatment: An FGFR-Related, but Growth Factor and Sex Steroid-Independent Mechanism?

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Keywords

FGFR inhibition · Erdafitinib · Growth promotion · Linear growth · Pediatric CNS tumors · Growth plate

Abstract

Introduction: Growth acceleration during postnatal growth only occurs during puberty as a physiological event and during catch-up growth mediated by growth-promoting therapies in growth disorders. Here we report on novel observations of skeletal symptoms during treatment with erdafitinib, a tyrosine kinase inhibitor (TKI) prescribed on the basis of a compassionate-use program. **Methods:** Analysis of anthropometric, biochemical, clinical, and radiographic data of patients with CNS tumors who revealed an unanticipated growth spurt with initiation of therapy with erdafitinib was performed retrospectively. **Results:** Linear growth acceleration was independent of sex steroids and IGF1 levels, which is especially remarkable in the context of heavily pretreated pediatric neuro-oncology patients with severe growth impairment before initiation of therapy. Growth acceleration was accompanied by a distinct widening of the growth plate and enhanced metaphyseal mineralization shortly after the start of TKI therapy. **Conclusions:** While targeted therapies including TKIs have become an essential part of adult cancer treatment, applications in children are still limited. Off-target effects specific to the

pediatric population have been observed in various organ systems; however, knowledge about the effect of TKIs on the growing skeleton is scarce. Treatment with erdafitinib inhibits FGFR3-mediated effects and thus represents a very logical hypothetical framework of growth factor and sex steroid-independent growth acceleration.

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Introduction

The growth plate, also known as the epiphyseal plate, is a region of developing cartilage tissue located near the ends of long bones in children and adolescents. It is responsible for linear growth and lengthening of bones by enchondral ossification, which involves the replacement of cartilage tissue with bone tissue. This process is orchestrated by a complex interplay of local

The findings have been presented as a poster presentation at the ESPE conference in Den Haag, The Netherlands, Sept 21st, 2023. A manuscript on the oncologic perspective of the included patients entitled "Feasibility and antitumour activity of the FGFR-inhibitor erdafitinib in 3 pediatric CNS tumor patients" has been published in *Pediatric Blood & Cancer* (<https://doi.org/10.1002/pbc.30836>).

regulators within the growth plate as well as systemic factors, such as GH and thyroid hormone. Pediatric neuro-oncology patients are particularly at risk for growth failure due to high exposure to chondrotoxic drugs, corticosteroids, and due to therapy-induced hypopituitarism [1].

Tyrosine kinase inhibitors (TKIs) target the activity of specific tyrosine kinases, enzymes that play a key role in distinct cell signaling pathways. TKIs have been approved for the treatment of several types of cancer in adults, including chronic myeloid leukemia, lung cancer, renal carcinomas, and gastrointestinal tumors [2]. Erdafitinib, a selective fibroblast growth factor receptor (FGFR) 1–4 TKI, has been approved for the treatment of patients with urothelial carcinoma and certain FGFR alterations and is currently further evaluated for patients with other tumors harboring FGFR alterations [3, 4]. FGFR3, a main target of erdafitinib, is a transmembrane receptor involved in the regulation of skeletal growth and development. Activating mutations in the FGFR3 gene, which is strongly expressed in growth plate chondrocytes, are associated with skeletal dysplasias such as achondroplasia [5]. Here, we report on linear growth acceleration associated with erdafitinib treatment in 2 patients without skeletal dysplasia.

Materials and Methods

In this report, we evaluate 2 pediatric patients with CNS tumors treated at the Medical University of Vienna. Medication was obtained on a compassionate use basis after exhaustion of standard therapy options and exclusion of study inclusion possibilities. Analysis was performed by retrospective in-depth data collection including print and/or digital patient charts, the local registry for growth disorders (Research Data Analysis, IT4Science, Medical University of Vienna), clinical summaries, as well as laboratory, radiological, histopathological, and genetic data. Anthropometric measurements were referenced to the Austrian reference dataset [6].

Results

Patient 1 was diagnosed with a mesencephalic glioma, NOS (not otherwise specified) grade °II–III at 5 years of age, with leptomeningeal metastases to the supratentorial ventricles. After first-line therapy consisting of focal irradiation and temozolomide therapy, growth hormone (GH) deficiency was diagnosed at the age of 8 years and GH treatment was initiated at initially 25 µg/kg/day (Fig. 1a). Thyroid function was normal at all timepoints. Bioptic reevaluation

at local progression in the brainstem at age 12 years revealed a low-grade tumor with an FGFR1 alteration. In lack of alternatives, off-label treatment with erdafitinib was initiated at 13.8 years. At this time, there was hypogonadotropic hypogonadism with unmeasurable LH, FSH, and testosterone serum levels, and the testicular volume of 6 mL was attributed to spontaneous puberty in the past followed by treatment-induced arrest. The initial erdafitinib dose of 5 mg/day had to be paused and lowered to 3 mg/d for bone pain and abdominal aches. Reevaluation after 6 months of erdafitinib revealed a dramatic growth spurt (Fig. 1a) without any evidence for pubertal progression, alterations in GH-treatment response, or BMI increase (Table 1; online suppl. Fig. S1; for all online suppl. material, see <https://doi.org/10.1159/000540485>). Wrist imaging revealed an atypical physal widening without apparent progression of bone maturation (Fig. 1c, d). Further, profound metaphyseal sclerosis was observed in association with the onset of erdafitinib treatment (Fig. 1c–f). Both erdafitinib and GH therapy were stopped due to the growth phenotype, and testosterone was initiated to advance pubertal development and induce growth plate fusion (Fig. 1a).

Patient 2 was diagnosed with an FGFR3-overexpressing ependymoma (PFA) at 4 years of age. There was no alteration of the endocrine function of therapeutic relevance. After standard therapy with irradiation and chemotherapy, she suffered from recurrences and tumor progression. Treatment was switched to erdafitinib at 10.9 years of age. With onset of treatment with erdafitinib, there was an unanticipated growth spurt despite previous growth retardation, prepubertal Tanner stage 1 with serum-estradiol below the detection limit, lack of BMI increase and low IGF-1 levels (−3.8 SDS, −3.5 SDS; Fig. 1b, online suppl. Fig. S1). Oncological aspects of erdafitinib treatment in the presented cases have been published recently [4].

Discussion

Growth acceleration during postnatal growth only occurs during puberty as a physiological event and during catch-up growth mediated by growth-promoting therapies in growth disorders. Here we describe the first cases of a pharmacologic induction of linear growth, independent of endogenous or exogenous GH, IGF-I, or sex-steroids, respectively.

While the first generation of FGFR inhibitors also targeted a broad spectrum of other kinases, the second generation including erdafitinib exhibits higher selectivity. Erdafitinib most specifically binds to FGFR1,

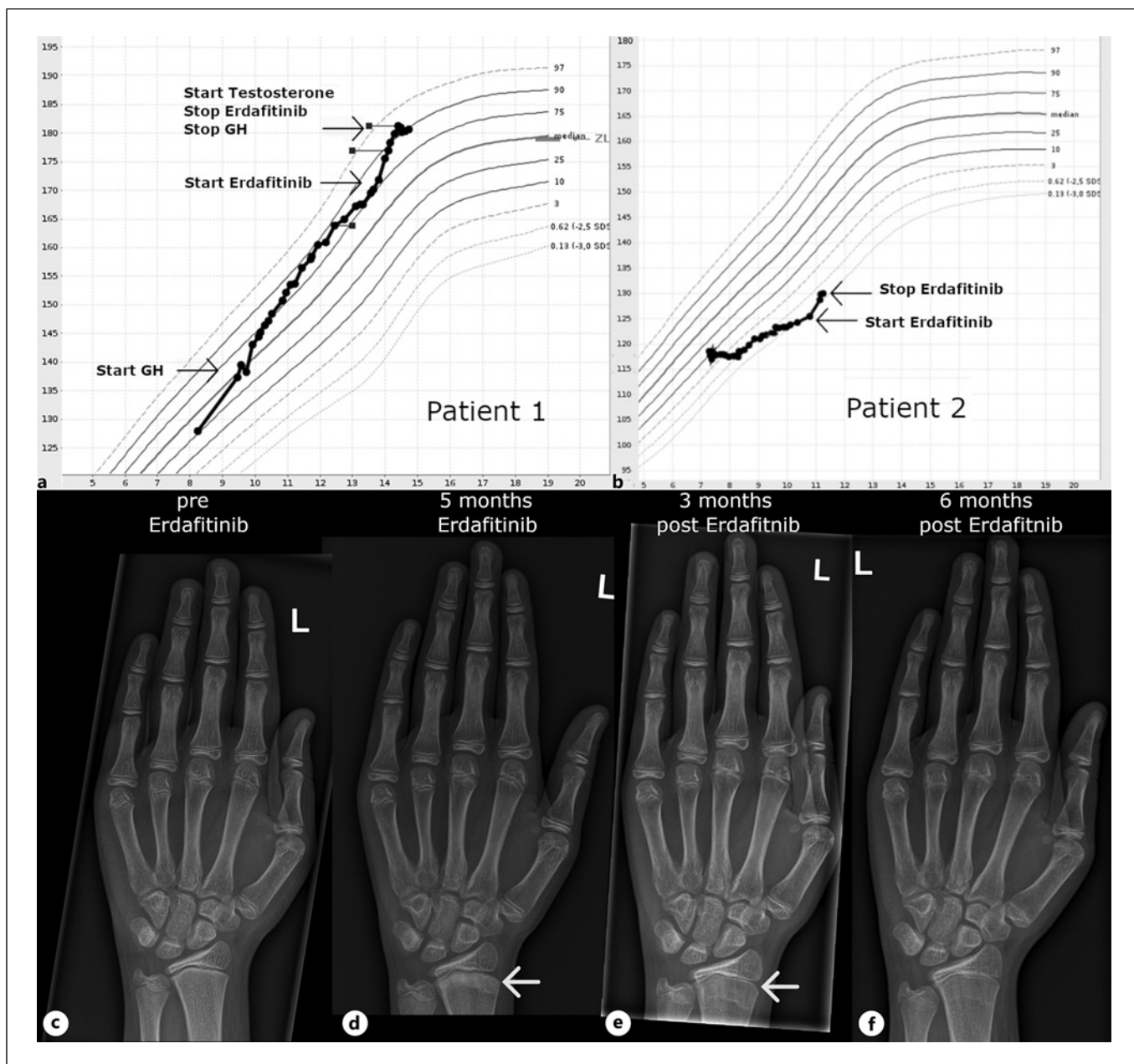


Fig. 1. Linear growth was accelerated in patient 1 (a) and patient 2 (b) in association with treatment onset of erdafitinib depicted on Austrian reference data [6]. Development of metaphyseal sclerosis in patient 1 during erdafitinib treatment with subsequent normalization of bone mineralization after treatment halt (c–f).

followed by FGFR3, 4, 2, and other kinases [7]. Interestingly, the growth acceleration in patient 2 did not occur during previous treatment with nintedanib, a triple angiokinase inhibitor, targeting VEGFR1-3, PDGFR- α and - β , and FGFR1-3 [8] (growth velocity nintedanib: 2.6 cm/year; growth velocity under erdafitinib 10 cm/year).

FGFR3, a main target of erdafitinib, is strongly involved in longitudinal growth as demonstrated by severely stunted growth in gain-of-function mutations of the FGFR3 leading to several human chondrodysplasias including achondroplasia [9]. Preclinical studies of pharmacological FGFR3 inhibition in mice showed an improvement in the length of the upper and lower

Table 1. Laboratory values in patients 1 and 2 during the course of erdafitinib treatment

Patient 1	Unit	Pre-erdafitinib	3 months erdafitinib	6 months erdafitinib	2 months post-erdafitinib	Reference range
Testosterone	ng/mL	<0.03	<0.03	<0.03	<0.03	<0.03 (Tanner 1)
IGF1	ng/mL	297	226	187	50	see SDS
IGF1	SDS	−0.04	−0.9	−1.4	−3.8 (↓)	−2.0–2.0
25OH-vitamin D	nmol/L	49.9 (↓)	67.8		57.5	50–250
Total ALP	U/L	191		341		116–468
Serum phosphate	mmol/L	2.47 (↑)	1.88 (↑)	2.01 (↑)	1.81	1.28–1.82
GH dosage	μg/kg/d	16	14	15	0*	
Testicular volume	mL	6	6	6	6	
Patient 2	Unit	Pre-erdafitinib	2 months erdafitinib	1 month post-erdafitinib	2 months post-erdafitinib	Reference range
Estradiol	pg/mL	<5		<5	<5	<5 (Tanner 1)
IGF1	ng/mL	20		21	45	see SDS
IGF1	SDS	−3.7 (↓)		−3.8 (↓)	−3.0 (↓)	−2.0–2.0
25OH-vitamin D	nmol/L	40.6 (↓)		56.3	84.3	50–250
Total ALP	U/L	86 (↓)		521 (↑)	438 (↑)	129–417
Serum phosphate	mmol/L	1.75	1.89 (↑)	1.88 (↑)	1.12 (↓)	1.28–1.82
Tanner stage		1	1	1	1	

*GH therapy was stopped 6 months after the start of erdafitinib.

extremities raising hopes for specific treatments including infigratinib, a selective FGFR1–3 TKI currently under investigation in children with achondroplasia [10].

In line with the described physiological function, loss of function of FGFR3 is associated with profound overgrowth in both mice and sheep [11, 12]. Further, an inactivating mutation of the FGFR3 tyrosine kinase domain was described in a patient with profound tall stature [13]. Taken together, experimental data in animal models as well as the first results of TKIs in the human model of FGFR3 in health and diseases point to a causative effect of FGFR3 inhibition by erdafitinib on the remarkable acceleration of growth observed in our 2 patients who do not suffer from skeletal dysplasia. Despite the observed association, the retrospective data analysis, the small number, and heterogeneous completeness of patients' datasets are clear limitations of this study. Further, the absence of an unknown influence or interaction of the GH treatment in patient 1 with the observed growth acceleration can only be assumed based on stable GH dose and IGF1-levels, but not ruled out due to the nature of data collection. Nevertheless, a substantial acceleration of linear growth velocity in critically ill pediatric patients without profound improvement of the underlying condition represents an unexpected finding requiring an evaluation of possible underlying mechanisms. In the context of the association

with novel pharmacological treatments, the reported findings aim to stimulate the prospective collection and growth-related focus of data collection in pediatric patients with TKI, particularly with erdafitinib treatment.

Conclusion

The increase in growth velocity associated with erdafitinib treatment observed in our patients points to a potent growth-promoting effect of FGFR3 inhibition. While similar strategies are under investigation in patients with FGFR3-activating mutations, we observed this effect in even higher magnitude in children without any suspected underlying abnormalities in skeletal FGFR3 function. The context of heavily pretreated pediatric cancer patients with severe growth impairment before initiation of erdafitinib underlines the hypothetical potential of a growth factor and sex steroid-independent growth acceleration by selective FGFR3-inhibition. This possible side effect has to be taken into account in the decision making for children and adolescents before the closure of the epiphyseal joints in need of oncologic treatment. Future studies will reveal if pharmacologic interventions on FGFR3 can complement current treatment strategies for short stature.

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Statement of Ethics

Written informed consent was obtained by the legal guardians, and the study was approved by the institutional review board of the Medical University of Vienna (1244/2016). In addition, written informed consent was obtained from the parent/legal guardian of the patient for publication of the details of their medical case and any accompanying images.

Conflict of Interest Statement

The authors have no relevant financial or non-financial interests to disclose.

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Adalbert Raimann is a consultant for Kyowa Kirin and received lecturing honoraria from Kyowa Kirin and Alexion. Johannes Gojo is a consultant for Roche and Novartis. Amedeo Azizi received

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Author Contributions

Adalbert Raimann, Natalia Stepien, Amedeo A. Azizi, Gabriele Hartmann, and Johannes Gojo contributed to the study conception and design. Data collection and analysis were performed by Adalbert Raimann, Natalia Stepien, Gabriele Hartmann, and Johannes Gojo. The first draft of the manuscript was written by Adalbert Raimann, Natalia Stepien, und Gabriele Hartmann, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data Availability Statement

The data that support the findings of this study are not publicly available since they contain identifying information about the study participants but are available from the corresponding author G.H. (gabriele.hartmann@meduniwien.ac.at) upon reasonable request.

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